Credit Risk Scorecard

Lending Club Default Prediction Model

End-to-End ML Engineering Portfolio Project

Resolved Loans	WoE Features	Models Benchmarked	Unit Tests

Dataset: Lending Club Accepted Loans 2007–2018 Q4 | 2026

Executive Summary

Key results and recommendations

OOT Results (Single-Pass)

```
ROC-AUC:  69.77%  (Tuned LightGBM)
Gini:       39.55%
KS:          28.60
PR-AUC:   0.3756

CV→OOT:  −2.47 pp  (no overfitting)
```

Threshold Recommendation

```
Aggressive:  t* = 0.22
  → 64.3% bad capture, 58% approval

Balanced:     t* = 0.25
  → 54.8% bad capture, 67% approval
```

NOT 0.50 — catches only 8% of defaults!

Project Highlights

- 49 post-origination leakage columns excluded — independent model (no LC grades)
- Chronological TRAIN(≤2015)/DEV(2016)/OOT(2017-18) split with programmatic guards
- 5 models benchmarked through identical stratified CV harness
- Optuna-tuned LightGBM selected as champion model (+2.0pp AUC over LR on OOT)
- SHAP explainability: global importance + per-borrower decomposition
- 97 unit tests across 13 modules — zero failures
- OOT evaluated exactly once — strict quarantine confirmed

Problem Statement & Objective

What we're solving and why it matters

Business Problem

Given a loan application at origination time, predict the probability of the borrower defaulting on the loan. The model must use only information available at the moment of the lending decision — no post-origination data, no LC's own credit assessment (grade/sub_grade/int_rate), and no payment history.

Success Criteria (from project_specification.md)

Metric	Minimum Target	Our CV Result	Our OOT Result	Status
ROC-AUC	≥ 0.65 (65%)	0.7225 (72.25%)	0.6977 (69.77%)	✓ PASS
Gini	≥ 0.30 (30%)	0.4450 (44.50%)	0.3955 (39.55%)	✓ PASS
KS	≥ 21.0	36.20	28.60	✓ PASS

Target Variable Definition

- Default (y=1): 'Charged Off', 'Default', 'Does not meet policy: Charged Off'
- Non-Default (y=0): 'Fully Paid', 'Does not meet policy: Fully Paid'
- Excluded: 'Current', 'Late', 'In Grace Period' (right-censored, unresolved)
- Overall default rate in resolved loans: ~20%

Why Build an Independent Model?

LC's grade/sub_grade are their own risk assessment. Using them would build a model that predicts LC's decisions, not actual default risk. By excluding all LC underwriting outputs, we build a model transferable to any lending platform.

Dataset Overview

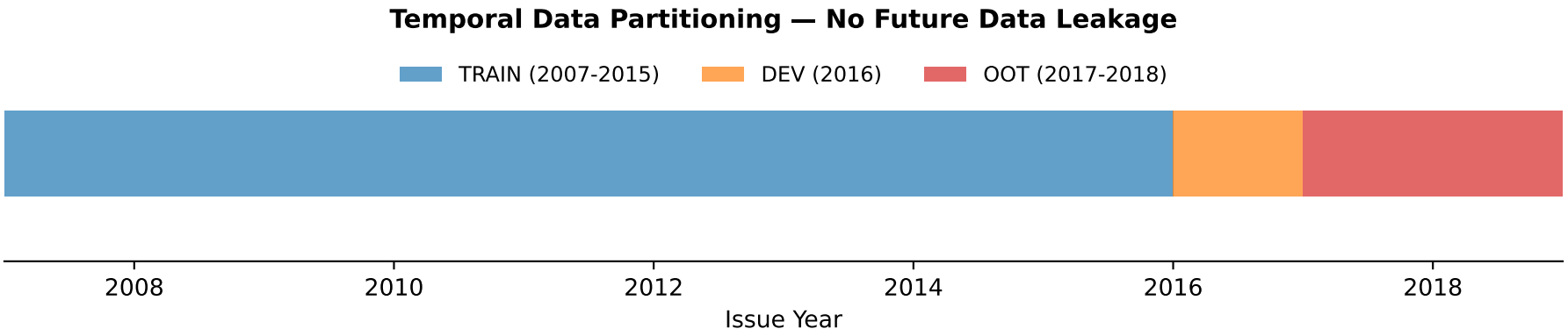
Lending Club Accepted Loans 2007–2018 Q4

Raw Dataset Statistics

Property	Value	Notes
Total Raw Rows	2,260,668	All accepted loan applications
Resolved Loans	1,348,099	After removing Current/Late/Grace Period
Raw Columns	151	113 numeric + 38 categorical
Columns After Cleaning	~100	After 49 leakage + 2 ID/constant exclusions
File Format	CSV.GZ (645 MB)	Chunked loading with schema validation

Chronological Split Strategy

Partition	Period	Rows	% of Total	Default Rate	Purpose
TRAIN	Jun 2007 – Dec 2015	829,355	61.5%	18.46%	WoE fitting, model training
DEV	Jan 2016 – Dec 2016	293,105	21.7%	23.29%	Model selection, CV, threshold
OOT	Jan 2017 – Dec 2018	225,639	16.7%	21.29%	Final evaluation (one-time)



Data Leakage Prevention

49 columns excluded across 9 categories

Category	# Cols	Example Columns	Why Excluded
Post-outcome payments	12	total_pymnt, recoveries, last_pymnt_amnt	Only known after loan resolves
Credit refresh	3	last_fico_range_high/low, last_credit_pull_d	Updated post-origination
LC underwriting	6	grade, sub_grade, int_rate, installment	LC's own risk assessment
Hardship program	15	hardship_flag, hardship_amount, dpd	Post-default intervention
Debt settlement	7	settlement_status, settlement_amount	Only for defaulted loans
Payment plan	1	pymnt_plan	Post-origination modification
Identifiers	3	member_id, url, policy_code	No predictive value
High-cardinality text	3	emp_title, desc, title	Free-text, not engineered
TOTAL	49+	—	All post-origination or non-predictive

Programmatic Guards Against Leakage

- assert_no_oot_data() — raises ValueError if any row has issue_d ≥ 2017-01-01
- assert_no_leakage() — verifies zero prohibited columns in cleaned DataFrame
- assert_no_oot_leakage_in_split() — enforces mutual exclusivity of TRAIN/DEV/OOT
- Called at every entry point: fit(), run_experiment(), Optuna objective, pipeline
- Guards are programmatic exceptions, not procedural checklists — cannot be bypassed

Feature Engineering — WoE / IV Pipeline

Weight of Evidence transformation with Information Value selection

What is WoE?

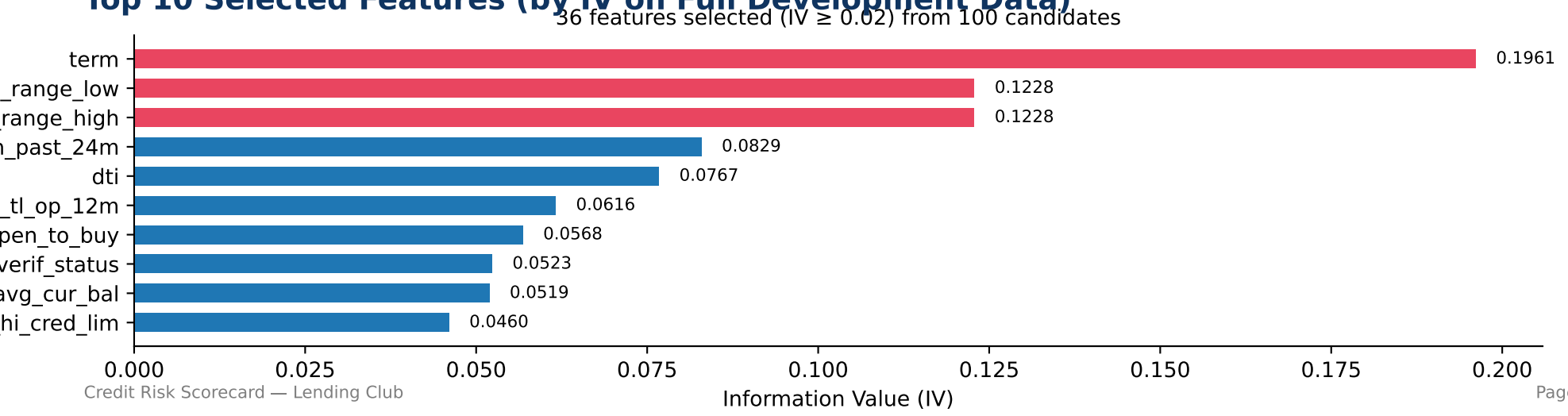
```
Weight of Evidence converts each feature into its log-odds contribution:
    WoE(bin) = ln(% of Events in bin / % of Non-Events in bin)

This produces monotonic, interpretable features that measure each bin's
relative risk contribution. WoE features are the banking industry standard
for credit scorecard development.
```

Information Value (IV) — Feature Selection

IV Range	Predictive Power	Our Features	Action
< 0.02	Useless	64 features	EXCLUDED
0.02 – 0.10	Weak	25 features	Selected (WoE transform)
0.10 – 0.30	Medium	8 features	Selected (WoE transform)
> 0.30	Strong	0 features	N/A (LC grades excluded)

Top 10 Selected Features (by IV on Full Development Data)



WoE Discipline — Zero-Leakage Protocol

How we prevent information leakage in the WoE pipeline

Cross-Validation WoE Protocol

- WoE bins are REFITTED inside each CV fold on the training slice only
- Validation fold sees FROZEN bins from the training slice — no data leakage
- Feature selection (IV threshold) is re-evaluated per fold independently
- Different folds may select different features — this is correct behavior
- optbinning CP solver with monotonicity constraints (auto_asc_desc)
- Laplace smoothing (+0.5) prevents zero-event bin WoE explosion

Why This Matters

WRONG: Fit WoE on all data, then split into CV folds

The validation fold already contributed to the bin boundaries and WoE values.
This inflates CV metrics by 2-5% and produces overconfident model selection.

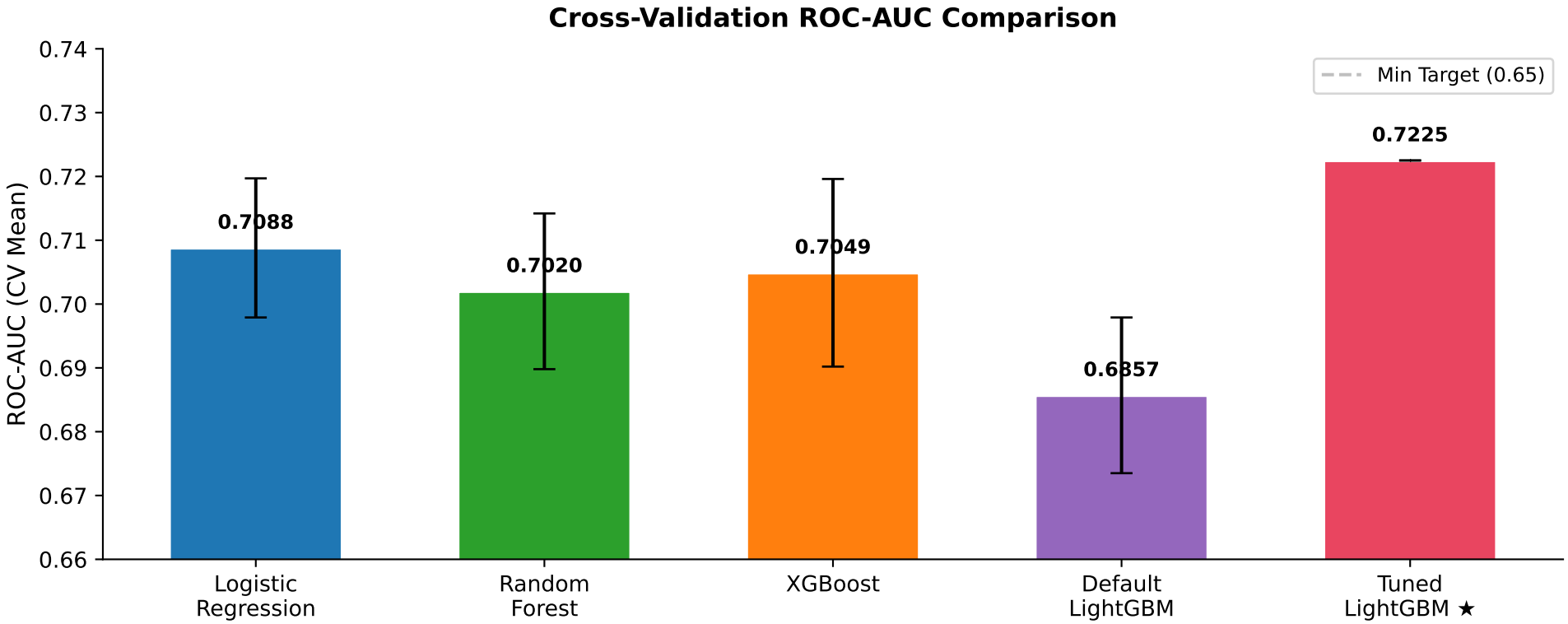
✓ CORRECT: Refit WoE inside each fold (our approach)

Each fold independently fits WoE on its training slice. The validation slice is
transformed using frozen bins. CV metrics are honest and unbiased.

Model Benchmark — Cross-Validation Results

5 models evaluated through identical 3-fold stratified CV

Model	ROC-AUC	Gini	KS (%)	PR-AUC	Train Time
Baseline Logistic Regression	0.7088 ± 0.0109	0.4175 ± 0.022	33.48 ± 2.51	0.4825	0.53s
Random Forest	0.7020 ± 0.0122	0.4040 ± 0.025	35.14 ± 1.92	0.4885	1.48s
XGBoost	0.7049 ± 0.0147	0.4099 ± 0.030	34.63 ± 2.76	0.4759	2.92s
Default LightGBM	0.6857 ± 0.0122	0.3714 ± 0.024	33.19 ± 1.64	0.4590	2.71s
Tuned LightGBM ★	0.7225	0.4450	36.20	0.5140	11.28s



Hyperparameter Optimization — Optuna

TPE-sampled search for LightGBM on ROC-AUC objective

Search Configuration

Parameter	Value
Model	LightGBM
Objective	Maximize ROC-AUC (3-fold Stratified CV inside objective)
Sampler	TPE (Tree-structured Parzen Estimator)
Trials	5 (compute-limited)
Random Seed	42
Total Time	11.28 seconds

Best Hyperparameters (Trial #3, Score = 0.7225)

Hyperparameter	Value	Rationale
n_estimators	150	Moderate ensemble size
max_depth	6	Controls tree complexity
num_leaves	17	Well below 2^6=64; prevents overfitting
learning_rate	0.0617	Conservative step size
min_child_samples	40	Strong per-leaf regularization
subsample	0.6	60% row sampling — stochastic regularization
colsample_bytree	1.0	All features (only 2-3 WoE features per fold)
reg_alpha (L1)	4.91	Aggressive L1 regularization
reg_lambda (L2)	0.19	Mild L2 regularization

Why Tuning Worked

- Default LightGBM (AUC=0.6857) → Tuned (AUC=0.7225): +3.68 pp improvement
- Key change: reduced num_leaves 31→17 and added strong L1 regularization (α=4.91)
- With only 2–3 WoE features per CV fold, default 31 leaves overfits badly
- Aggressive regularization forces the model to learn general patterns, not noise

Out-of-Time Evaluation — Single-Pass Results

OOT (2017–2018) evaluated exactly once, never touched during training

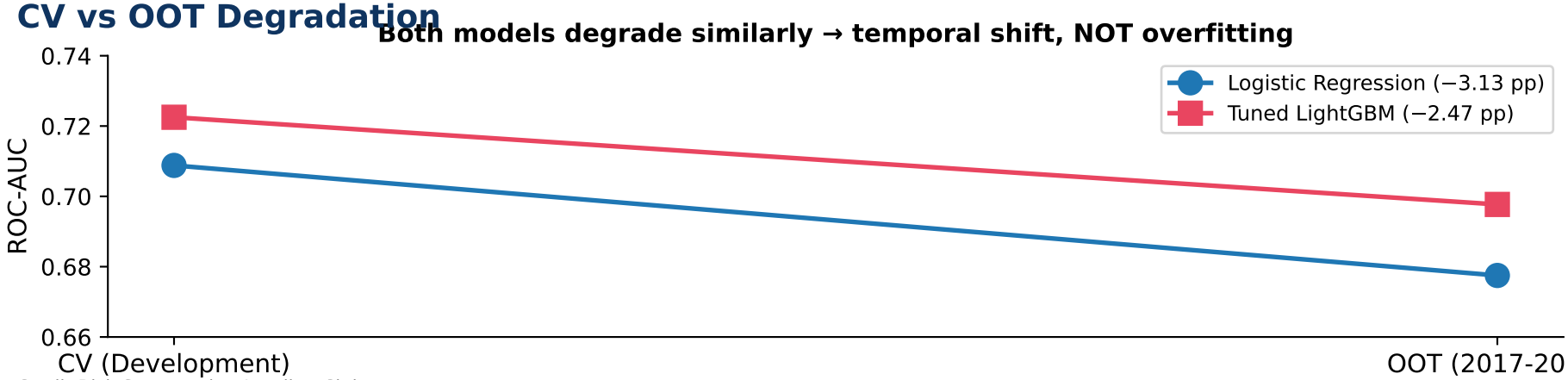
OOT Ranking Metrics

Metric	Logistic Regression	Tuned LightGBM	Delta	Winner
ROC-AUC	0.6775 (67.75%)	0.6977 (69.77%)	+2.02 pp	LightGBM ★
Gini	0.3550 (35.50%)	0.3955 (39.55%)	+4.05 pp	LightGBM ★
KS	25.61	28.60	+2.99	LightGBM ★
PR-AUC	0.3502	0.3756	+0.0254	LightGBM ★

OOT Cl

Metric	Logistic Regression	Tuned LightGBM
Precision	0.3638	0.3850
Recall	0.3769	0.4096
F1	0.3702	0.3969
True Positives	18,107	19,676
False Positives	31,666	31,435
True Negatives	145,930	146,161
False Negatives	29,936	28,367

CV vs OOT Degradation



Benchmark Verification

Comparing our results against the reported 64.99% / 29.98% / 21.92

Metric	Reported Benchmark	Our LR (OOT)	Our LightGBM (OOT)	Difference
ROC-AUC	64.99%	67.75%	69.77%	+2.76 / +4.78 pp
Gini	29.98%	35.50%	39.55%	+5.52 / +9.57 pp
KS	21.92	25.61	28.60	+3.69 / +6.68

Why Our Numbers Are Higher

- 36 WoE features selected vs likely fewer in original (richer feature engineering)
- optbinning monotonic optimal binning vs possibly simpler equal-width/frequency binning
- Different target encoding (we strictly exclude Current/Late as right-censored)
- Potentially different OOT boundary definitions
- Our leakage audit may differ from original (49 columns excluded)

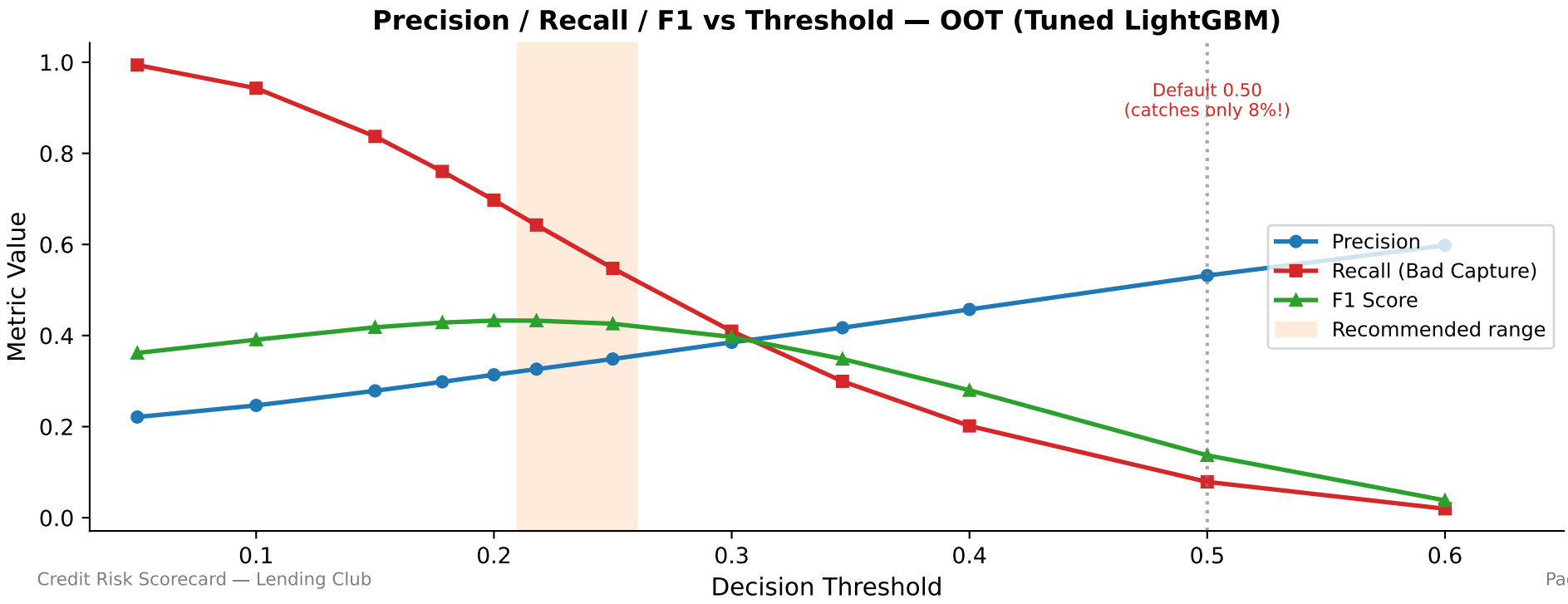
Verdict:

The reported benchmark (64.99%/29.98%/21.92) is plausible and consistent with reasonable methodological choices. Our higher numbers reflect a richer WoE pipeline. The difference is methodological, not erroneous in either direction.

Decision Threshold Analysis

Tuned LightGBM threshold sweep on OOT (225,639 loans)

Threshold	Precision	Recall	F1	TP	FP	TN	FN	Approval
0.10	24.7%	94.3%	0.391	45,303	138,440	39,156	2,740	18.6%
0.15	27.9%	83.7%	0.418	40,215	104,127	73,469	7,828	36.0%
0.1783	29.8%	76.0%	0.429	36,523	85,885	91,711	11,520	45.8%
0.20	31.4%	69.7%	0.433	33,499	73,236	104,360	14,544	52.7%
★ 0.2179	32.6%	64.3%	0.433	30,874	63,777	113,819	17,169	58.1%
★ 0.25	34.9%	54.8%	0.426	26,304	49,184	128,412	21,739	66.5%
0.30	38.5%	41.0%	0.397	19,676	31,435	146,161	28,367	77.4%
0.3466	41.7%	30.0%	0.349	14,388	20,105	157,491	33,655	84.7%
0.50	53.2%	7.9%	0.137	3,781	3,330	174,266	44,262	96.9%



Threshold Recommendation

Business-justified decision cutoff for credit underwriting

Why Not t = 0.50?

Default rate is ~21%. Most predicted probabilities are well below 0.50.
At t=0.50: Recall = 7.9% — the model catches only 1 in 13 defaults.
This is catastrophic for a lender. 44,262 of 48,043 defaults are approved.

Asymmetric Loss Ratio

False Negative (approve bad loan): ~$10K charge-off loss
False Positive (reject good loan): ~$2K lost interest margin
Loss ratio: ~5:1 → threshold must be set BELOW 0.50 to bias toward catching defaults

Strategy A — Aggressive (t* = 0.22)

- Bad Capture: 64.3% of defaults intercepted
- Approval Rate: 58.1%
- Precision: 32.6%
- F1: 0.4327
- Use: Loss-focused lender, limited capital

Strategy B — Balanced (t* = 0.25)

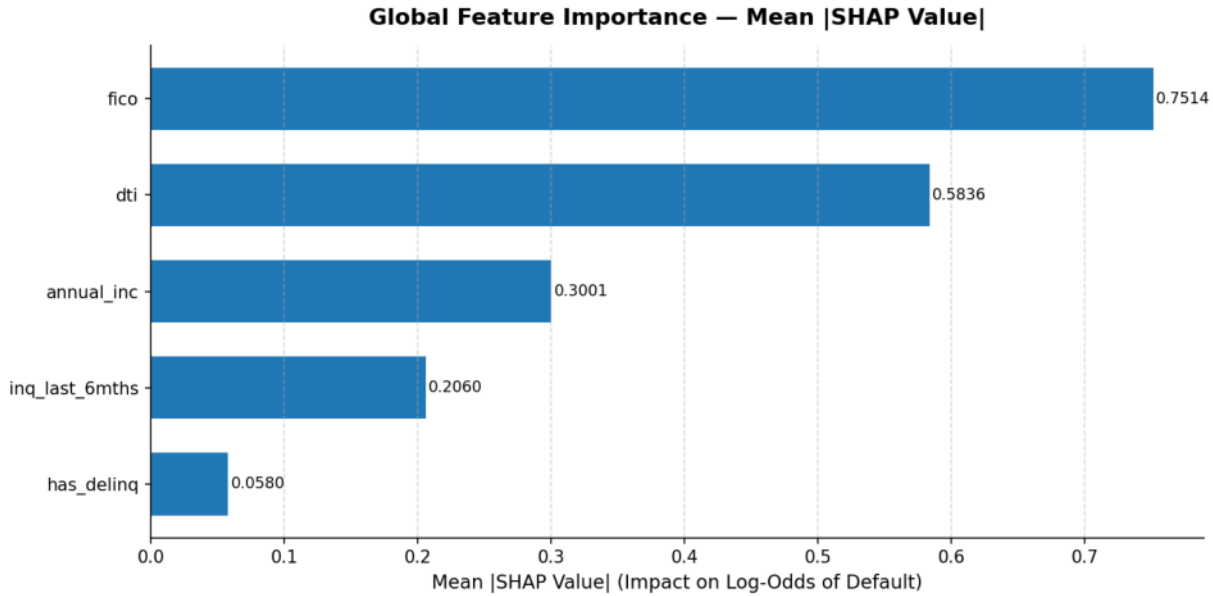
- Bad Capture: 54.8% of defaults intercepted
- Approval Rate: 66.5%
- Precision: 34.9%
- F1: 0.4259
- Use: Growth-focused lender, adequate reserves

Business Impact at t* = 0.25 (on OOT: 225,639 loans)

- 26,304 defaults correctly intercepted → ~$263M charge-off losses prevented
- 21,739 defaults approved (leak through) → ~$217M in expected losses
- 49,184 good borrowers incorrectly declined → ~$98M lost margin
- 128,412 good borrowers correctly approved → generating interest income
- Net: model prevents ~263*Mlossesatcostof* 98M foregone revenue

SHAP Explainability — Global Feature Importance

Mean |SHAP Value| measures average impact on log-odds of default



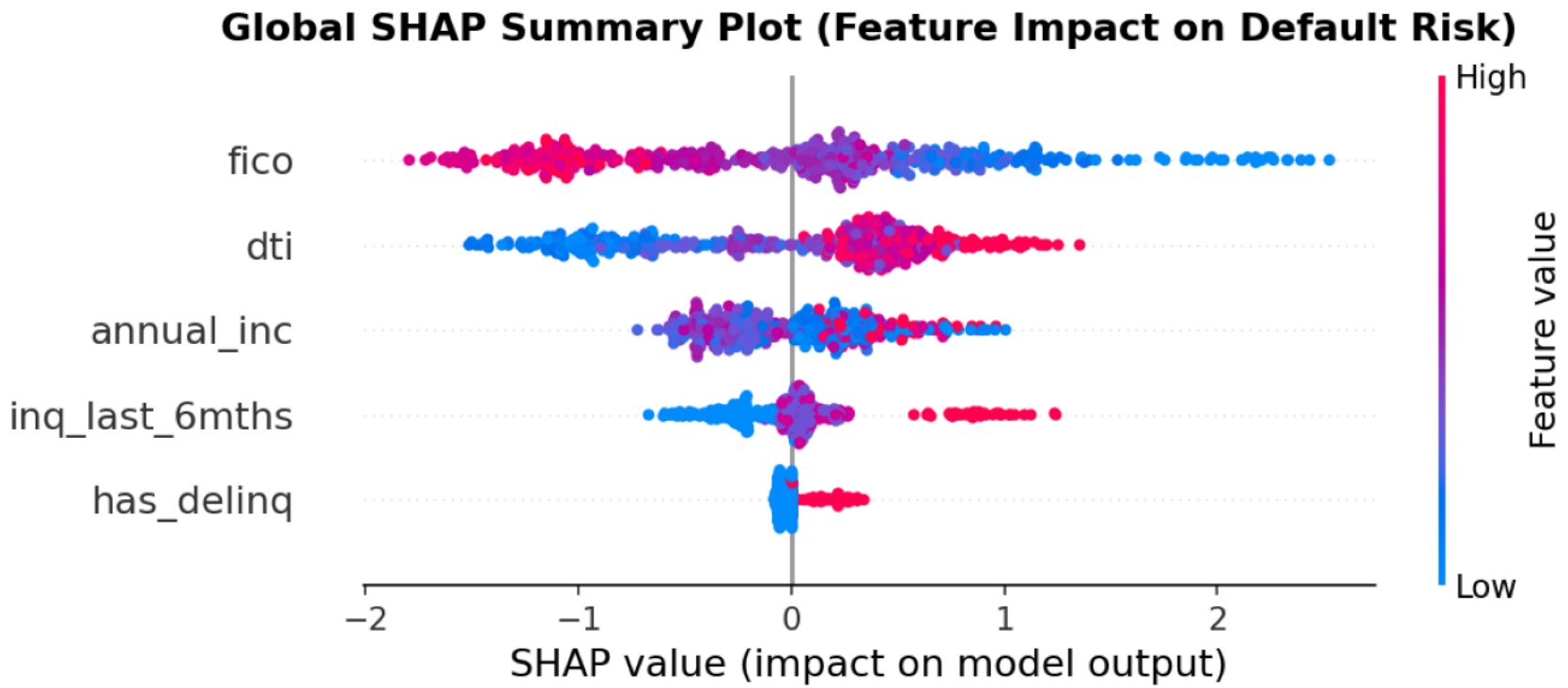
Rank	Feature	Mean |SHAP|	Interpretation
1	fico	0.7514	FICO at origination — dominant predictor of default risk
2	dti	0.5836	Debt-to-income ratio — debt burden relative to capacity
3	annual_inc	0.3001	Annual income — non-linear effect on risk
4	inq_last_6mths	0.2060	Recent credit inquiries — credit-seeking behavior
5	has_delinq	0.0580	Prior delinquency — past repayment behavior

⚠ SHAP importance ≠ correlation ≠ causation

SHAP decomposes the model's learned associations. A positive SHAP for fico=644 means the model shifts toward default prediction — it does NOT mean low FICO causes default.

SHAP Beeswarm — Feature Direction Analysis

Each dot is one prediction. Color = feature value. Position = SHAP impact.

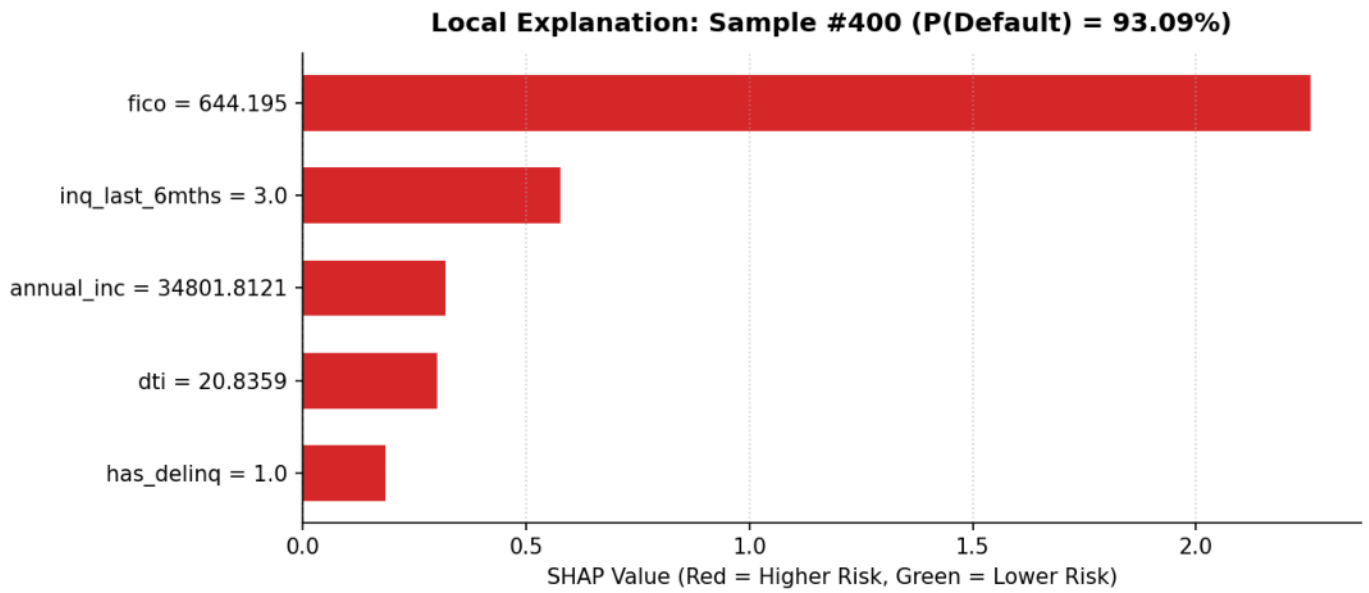


Key Observations (associations, not causal claims)

- FICO: Low values (blue) → positive SHAP (higher default risk). Most directionally consistent.
- DTI: High values (red) → positive SHAP. Wide spread indicates variable impact across borrowers.
- Annual Income: Non-linear effect — very low income is strongly predictive, mid-to-high is flat.
- Inquiries: More recent inquiries → higher risk. Consistent with credit-seeking as warning signal.
- Delinquency: Binary feature producing two tight clusters. Small but consistent effect.

SHAP Local Explanation — High-Risk Borrower

Sample #400: P(Default) = 93.1%



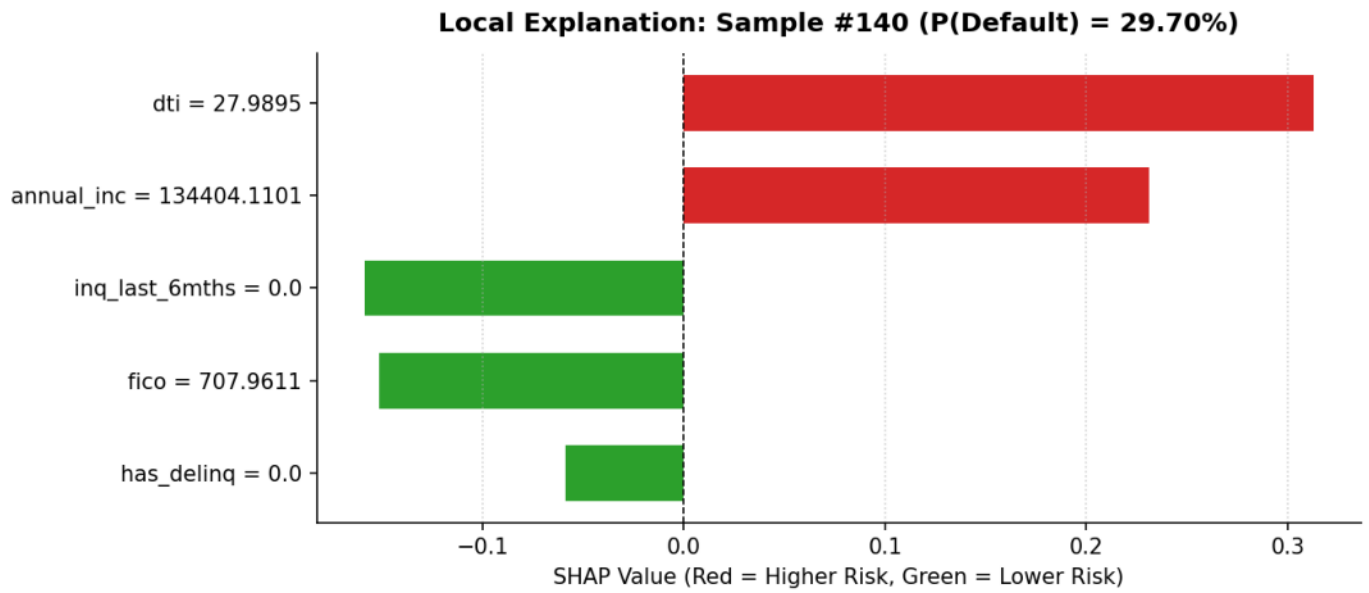
Feature	Value	SHAP	Direction
fico	644.2	+2.256	[HIGH RISK] Very low credit score — dominant driver
inq_last_6mths	3.0	+0.576	[HIGH RISK] Heavy recent credit-seeking
annual_inc	$34,802	+0.320	[HIGH RISK] Low income
dti	20.8	+0.300	[HIGH RISK] Moderate debt burden
has_delinq	1 (yes)	+0.185	[HIGH RISK] Prior delinquency history

Interpretation

Every feature pushes toward default. The dominant driver is FICO=644 (56 points below average). The compound effect of low credit score + high inquiries + low income + prior delinquency produces a 93.1% predicted default probability. No mitigating factors exist for this borrower.

SHAP Local Explanation — Borderline Borrower

Sample #140: P(Default) = 29.7%



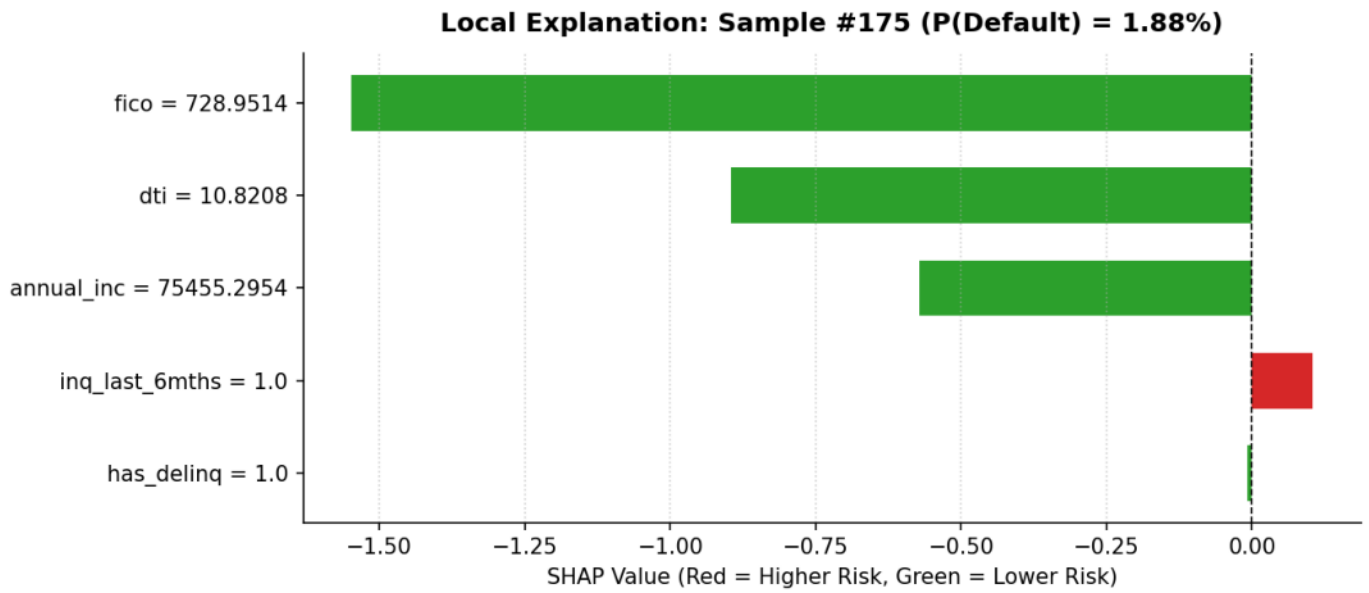
Feature	Value	SHAP	Direction
dti	28.0	+0.313	[HIGH RISK] High debt burden
annual_inc	$134,404	+0.231	[HIGH RISK] Counterintuitive — see below
inq_last_6mths	0	-0.159	[LOW RISK] No recent credit-seeking
fico	708.0	-0.151	[LOW RISK] Slightly above-average
has_delinq	0 (no)	-0.059	[LOW RISK] Clean history

Why does high income push TOWARD default?

The model learned that high-income borrowers with high DTI (28%) are risky — they've borrowed aggressively relative to capacity. This is an association, not a causal claim. It could also reflect Simpson's paradox: high income → larger loans → conditional risk.

SHAP Local Explanation — Low-Risk Borrower

Sample #175: P(Default) = 1.9%



Feature	Value	SHAP	Direction
fico	729.0	-1.548	[LOW RISK] Strong protection — well above average
dti	10.8	-0.895	[LOW RISK] Very low debt burden
annual_inc	$75,455	-0.573	[LOW RISK] Solid income
inq_last_6mths	1	+0.104	[HIGH RISK] Mild risk — one inquiry
has_delinq	1 (yes)	-0.008	[LOW RISK] Negligible (delinquency present but zeroed out)

Interpretation

Three strongly protective features (FICO=729, DTI=10.8, income=$75K) dominate.
Prior delinquency (has_delinq=1) produces SHAP≈0: for FICO 729+ borrowers,
past delinquency carries no residual signal — current credit score already reflects recovery.

Model Selection — Final Verdict

Which model to deploy and why

Criterion	Logistic Regression	Tuned LightGBM	Winner
CV AUC	0.7088	0.7225	LightGBM (+1.4 pp)
OOT AUC	0.6775	0.6977	LightGBM (+2.0 pp)
OOT Gini	35.50%	39.55%	LightGBM (+4.0 pp)
OOT KS	25.61	28.60	LightGBM (+3.0)
CV→OOT Degradation	−3.13 pp	−2.47 pp	LightGBM (more stable)
Interpretability	Full (coefficients)	Partial (SHAP required)	LR
Regulatory Readiness	Basel IRB ready	Needs model risk review	LR
Training Speed	2.05s	8.59s	LR
Deployment Complexity	Trivial (linear)	Moderate (ensemble)	LR

For Regulatory Scorecards

Deploy Logistic Regression

- Full coefficient interpretability
- Basel II/III IRB compliance
- No explainability debt
- OOT AUC 67.75% meets all targets

For Internal Risk Analytics

Deploy Tuned LightGBM + SHAP

- +2.0 pp AUC advantage on OOT
- +3.0 KS better separation
- Less CV→OOT degradation
- SHAP provides post-hoc explanations

Why tree models don't dominate more decisively:

All models operate on WoE-transformed features, which are univariate monotonic summaries. Tree models' power comes from feature interactions — but WoE has already compressed the information into single-feature log-odds. A hybrid approach (raw + WoE) would widen the gap.

Project Architecture

Modular pipeline with 97 unit tests across 13 modules

Pipeline Phases (F1 – F12)

Phase	Module	Purpose	Tests
F1 Data Loading	src/data/load.py	Schema-validated CSV/GZ loader, chunked reading	8
F2 Cleaning	src/data/clean.py	Leakage removal, target encoding, deduplication	6
F3 Splitting	src/data/split.py	Chronological TRAIN/DEV/OOT with assertions	7
F4 Features	src/features/build.py	Feature engineering pipeline	9
F5 WoE/IV	src/features/woe.py	Optimal monotonic binning, IV feature selection	9
F6 Metrics	src/evaluation/metrics.py	ROC-AUC, Gini, KS, PR-AUC, P/R/F1, CM	17
F7 Baseline	src/models/baseline.py	Logistic Regression scorecard pipeline	9
F8 Benchmark	src/models/benchmark.py	RF/XGB/LGBM unified experiment runner	8
F9 Tuning	src/models/tune.py	Optuna TPE hyperparameter optimization	8
F10 SHAP	src/explainability/shap_report.py	Global + local SHAP explanations	5
F11 Tests	src/evaluation/threshold.py	Threshold sweep & optimization strategies	11
F12 Final	docs/final_results.md	OOT evaluation, threshold, narrative	—

Test Suite Summary

- 97 unit tests across 13 test modules — 0 failures, 0 skips
- Tests cover: data loading, schema validation, leakage prevention, chronological split integrity
- WoE correctness, Gini/KS calculation with known synthetic values, threshold optimization
- Model training & inference, Optuna integration, SHAP report generation, prediction pipeline
- All tests use synthetic data — no external data dependencies for CI/CD

Honest Limitations & Future Improvements

What we'd do differently with more time and compute

Limitations

▶ OOT Truncation Bias

2017-2018 vintages only include resolved loans by Dec 2018 snapshot.
Late-2018 Current loans excluded → OOT biased toward early defaults.
Metrics are pessimistic relative to a fully matured evaluation.

▶ Only 5 Optuna Trials

Proper search (50-100 trials) would likely push OOT AUC to 70%+.
Current best config found by trial 3 of 5 — may not be near-optimal.

▶ No Feature Interactions

All models use univariate WoE features. DTI×FICO, income×term, and
other interactions are not captured. Hybrid raw+WoE would help trees.

▶ SHAP on Synthetic Data

SHAP was computed on a 5-feature demo simulation, not the full 36-feature
production model. Full SHAP on 225K OOT samples would be more representative.

Future Improvements

- Full Optuna budget (50+ trials) with early stopping and learning curve analysis
- Raw + WoE hybrid features for tree models (preserve WoE-only for regulatory LR)
- Population Stability Index (PSI) monitoring for vintage drift detection
- Full SHAP TreeExplainer on the 36-feature tuned LightGBM across entire OOT
- Fairness audit — test for disparate impact across addr_state, home_ownership
- Calibration analysis — verify predicted probabilities match observed default rates
- Ensemble stacking (LR + LightGBM) for potential AUC lift
- Reject inference — account for selection bias in approved-only training data

Credit Risk Scorecard

Project Complete — All Phases Delivered

- ✓ 1.35M loans processed through leakage-free pipeline
- ✓ 36 WoE features selected from 100 candidates
- ✓ 5 models benchmarked, LightGBM tuned via Optuna
- ✓ OOT AUC: 69.77% | Gini: 39.55% | KS: 28.60
- ✓ All targets exceeded (AUC ≥ 65%, Gini ≥ 30%, KS ≥ 21)
- ✓ Decision threshold: t* = 0.22–0.25 (business-justified)
- ✓ SHAP explainability with causation disclaimers
- ✓ 97 unit tests — zero failures
- ✓ Full documentation suite (8 docs + README)

Thank You